

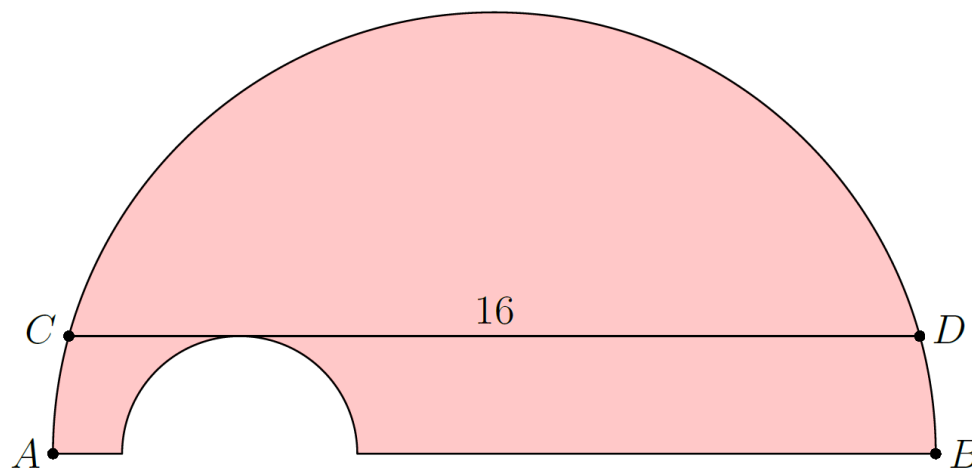
2025 AMC 10A Problem 10 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 10. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 10

Problem:

A semicircle has diameter $\overline{AB}$ and chord $\overline{CD}$ of length 16 parallel to $\overline{AB}$. A smaller semicircle with diameter on $\overline{AB}$ and tangent to $\overline{CD}$ is cut from the larger semicircle, as shown below.



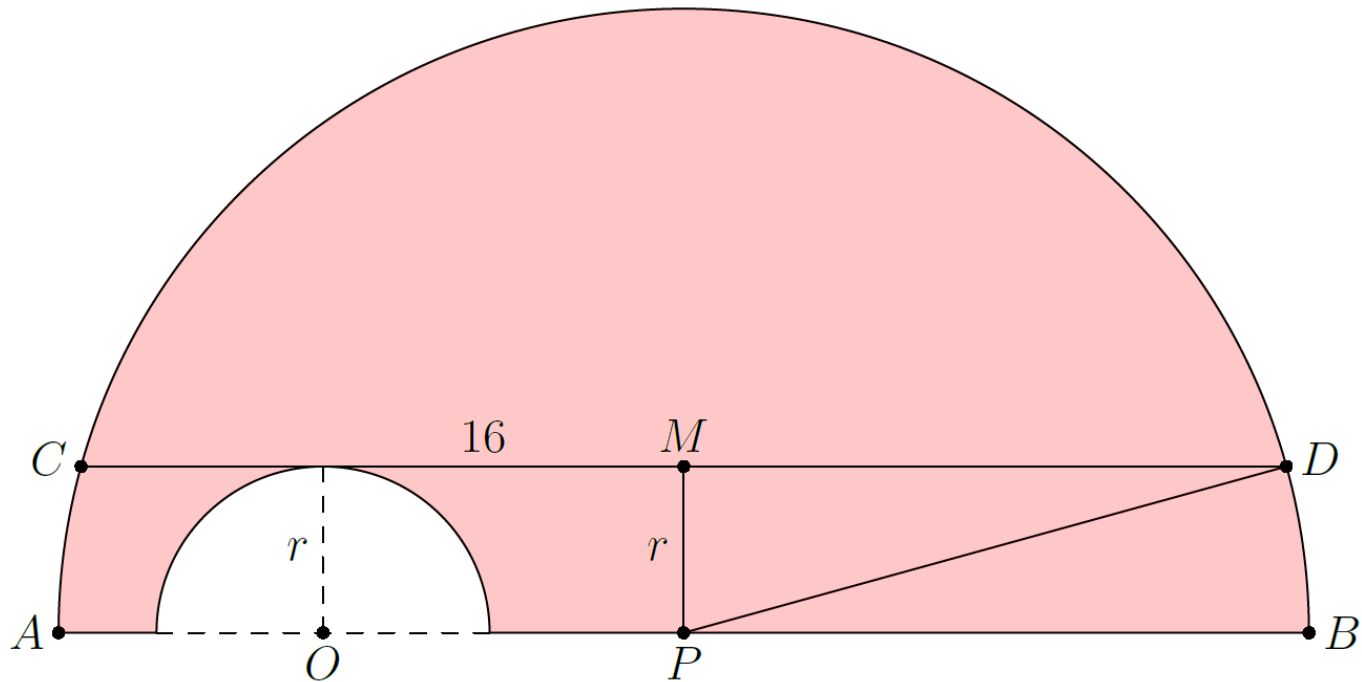
What is the area of the resulting figure, shown shaded?

Answer Choices:

A. 16π

B. 24π

C. 32π

D. 48π E. 64π **Solution:**

Let the radius of the small circle with center O be r , as shown in the diagram. Let P be the center of the larger (pink) semicircle with radius R . Let M be the midpoint of the chord CD . Since M is the midpoint of CD , we have

$$PM \perp MD.$$

From the figure,

$$PM = r, \quad PB = R, \quad \text{and} \quad MD = 8.$$

Applying the Pythagorean Theorem in $\triangle PMD$,

$$R^2 - r^2 = 8^2 = 64.$$

The required shaded area (the large semicircle minus the small semicircle) is

$$\frac{\pi(R^2 - r^2)}{2} = \frac{\pi \times 64}{2} = 32\pi.$$

which gives us the answer

$$\boxed{(C) \ 32\pi}$$

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